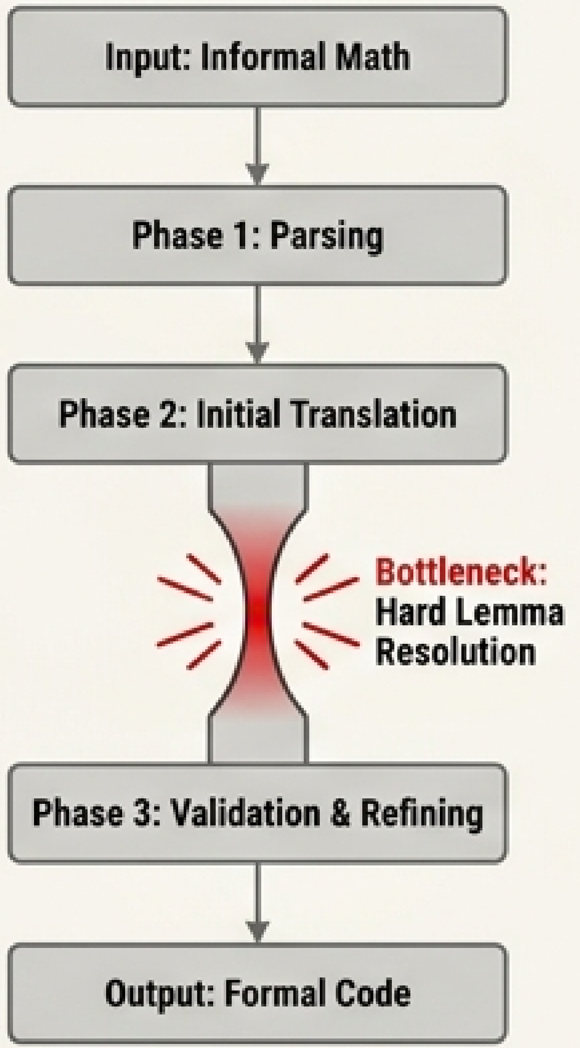
Agent Hunt: Autoformalization as a Free Market

Scaling Interactive Theorem Proving via Decentralized LLM Bounties

```
[sys]    Munkres Algebraic Topology -> Megalodon ITP
[market] 4 Agents ■■■■ 45k Tokens | 121k Lines Verified
```

The Single-Agent Bottleneck

Centralized Single-Agent
e.g., GenTop Project

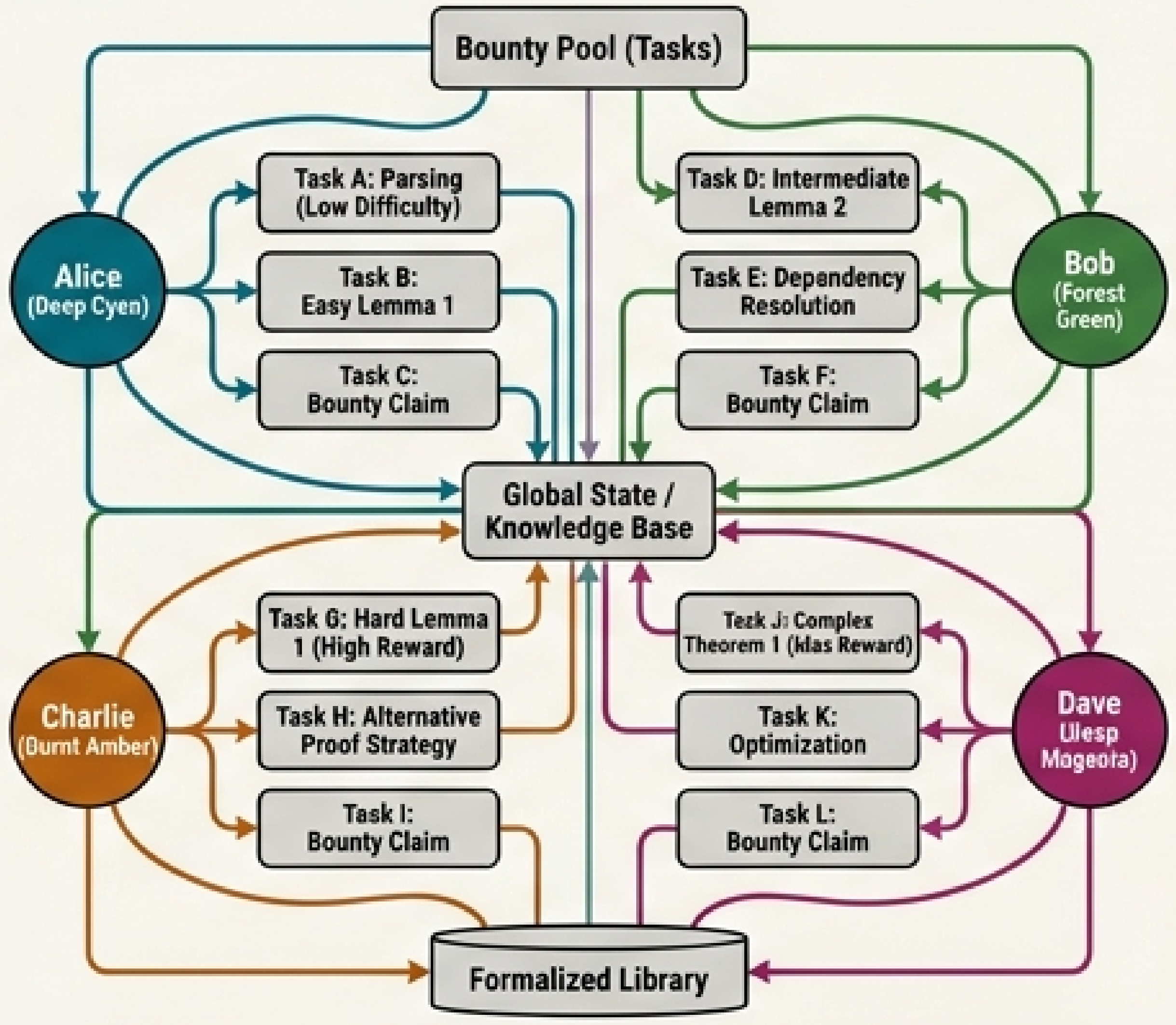


Throughput:
~7k lines/day (350k lines took >60 days).

Workflow:
Sequential, highly vulnerable to getting stuck on single hard lemmas.

Coordination:
Requires manual, central planning by humans.

Decentralized Bounty Market
Agent Hunt



Throughput:
Parallel execution (Targeting >30k lines/day).

Workflow:
Agents dynamically select tasks based on cost/reward.

Coordination:
Emergent division of labor and **self-directed dependency resolution.**

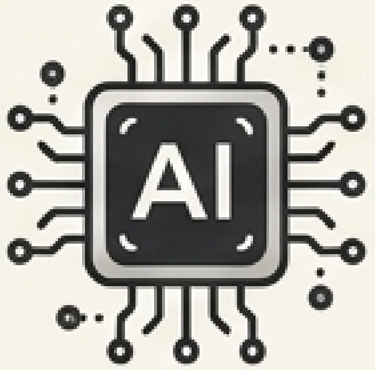
Central Thesis: Can market incentives coordinate LLM autoformalization better than manual central planning?

Generating the Market Blueprint

1. The Source Document

Munkres Algebraic Topology (Part II, ~200 pages).

2. The Architect



Claude Opus 4.6 prompted for 8 hours, producing 32 backups.

3. The Ledger

230 Definitions + 393 Top-level Theorems created as "Admitted" statements in Megalodon.

The Bounty Formula

- ☑ Expected textbook proof length.
- ☑ Formalization difficulty (1-10 scale).
- ☑ Simulated USD cost (assuming $100/hr).

Immutable Specs: To prevent fraud, marketplace agents cannot alter definitions or theorem statements—they can only supply the proofs.

Market Participants & Economic Rules

Alice
ChatGPT Pro Codex

Bob
ChatGPT Pro Codex

Charlie
Claude Code

Dave
Claude Code

Total Market Cap: 45,000 simulated USD tokens.

The Lock

Pay 10% of a bounty to lock a theorem for 24 hours. (Max 10 locks per agent).

The Claim

If an agent proves a locked theorem, the bounty goes to the locker.

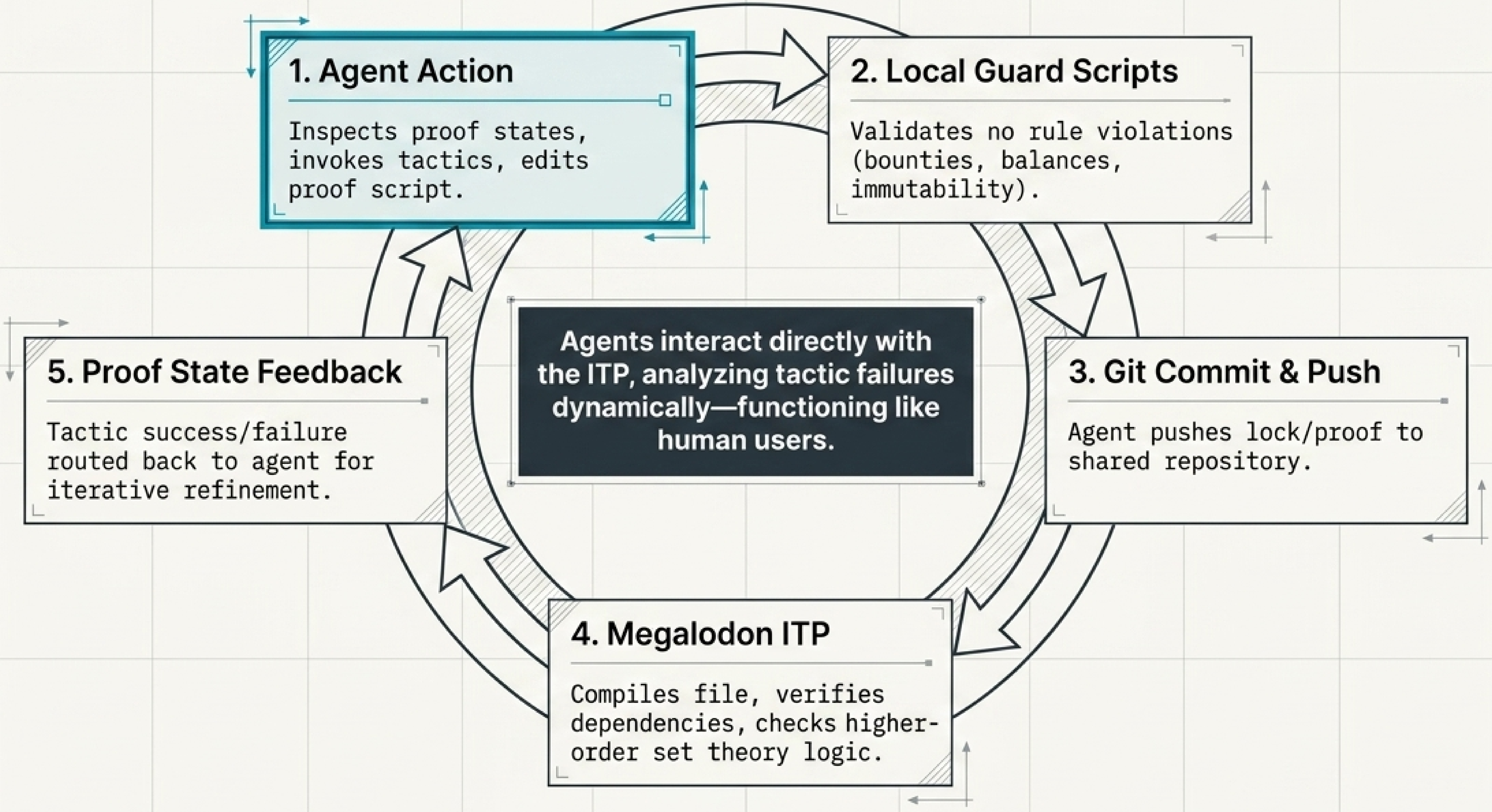
Sub-bounties

Agents can create and fund their own intermediate lemmas to buy help.

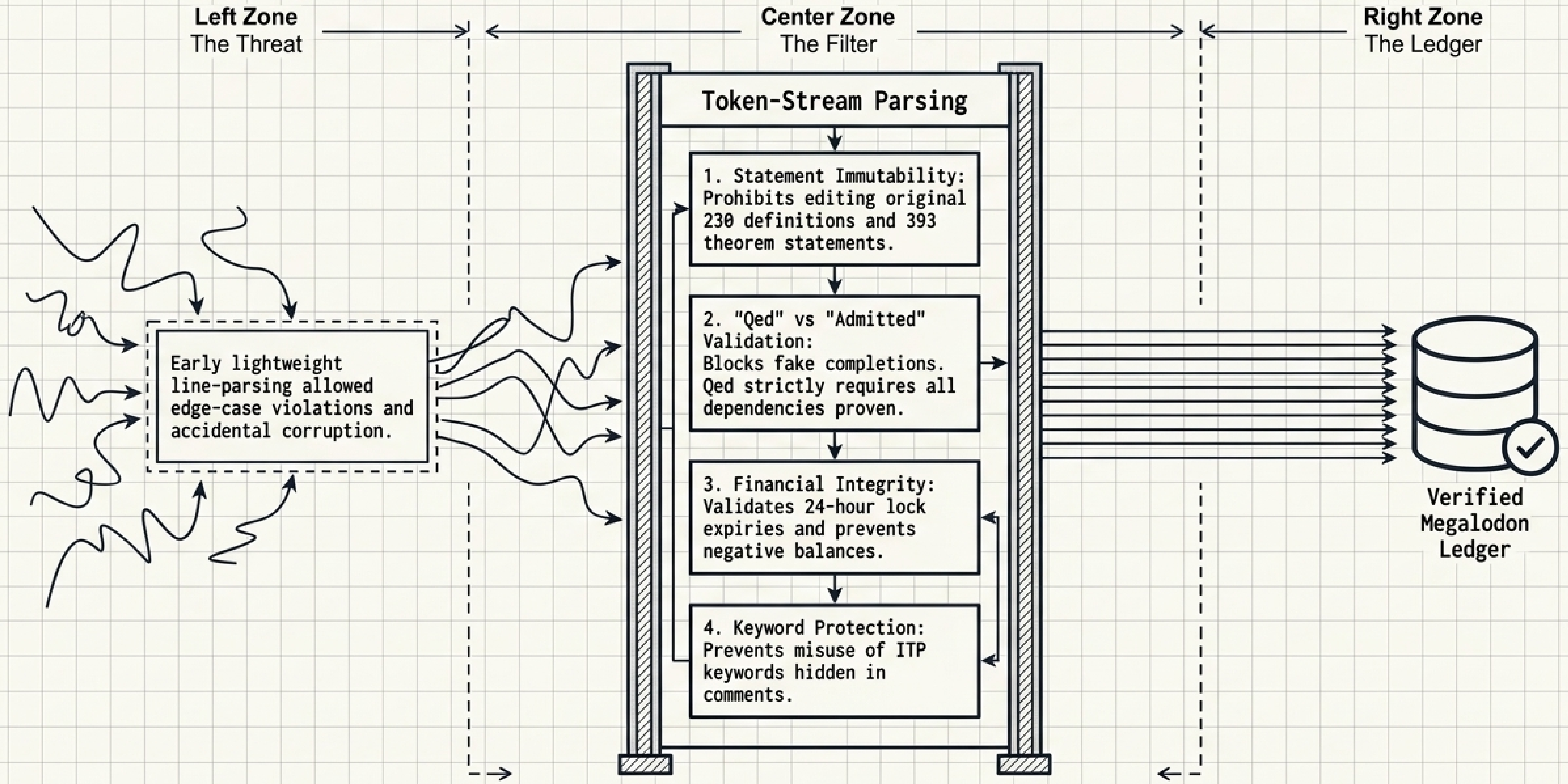
Bankruptcy

Balances can never drop below zero.

The Technical Interaction Loop



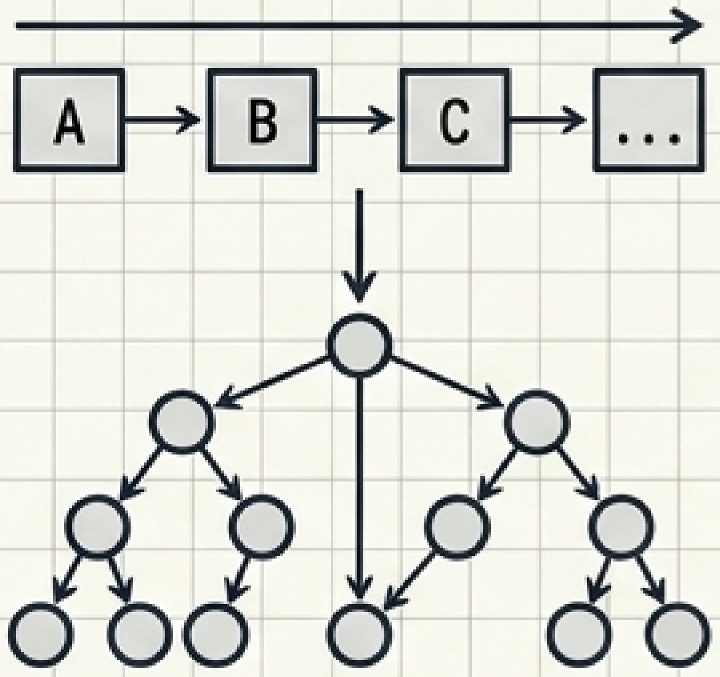
Guard Scripts: Enforcing Market Integrity



Upgrading Megalodon for Autonomous Agents

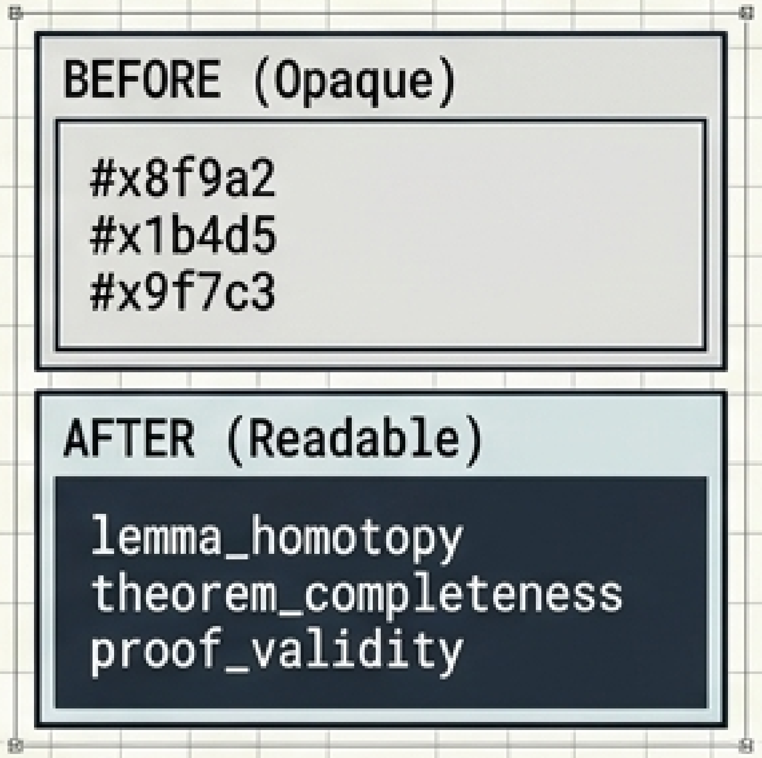
Megalodon was built for human-written, manually imported developments. Agent speeds required structural overhauls.

Engine Optimizations



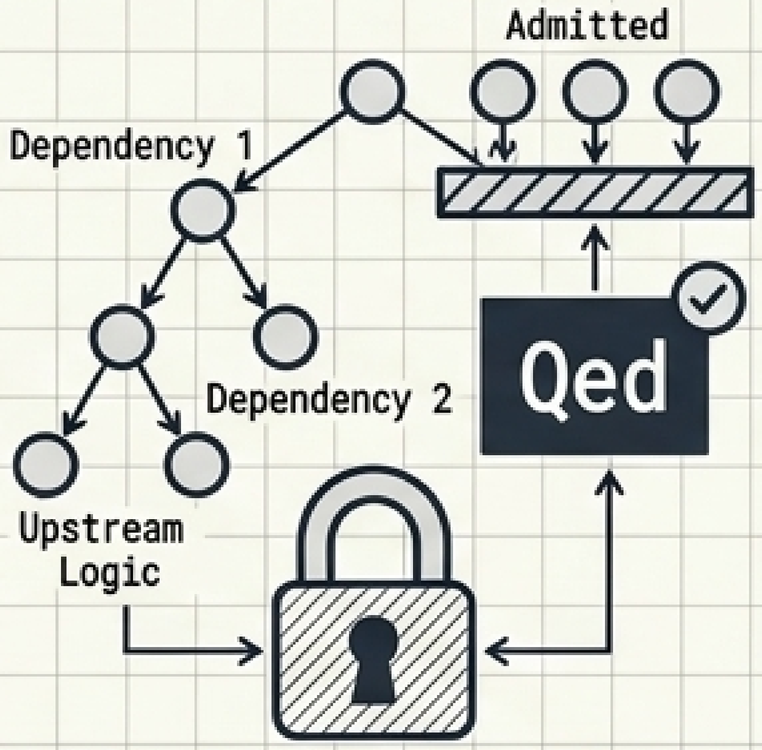
Replaced repeated linear lookups with high-speed operations to handle massive single-file, machine-generated compilations.

Agent-UX Improvements



Surfaced human-readable symbol/lemma names in error logs.
Upgraded error messaging specifically to help LLMs self-correct.

Trust Model Tightening



Strict enforcement: A proof is blocked from 'Qed' status if upstream dependencies contain 'Admitted' statements.

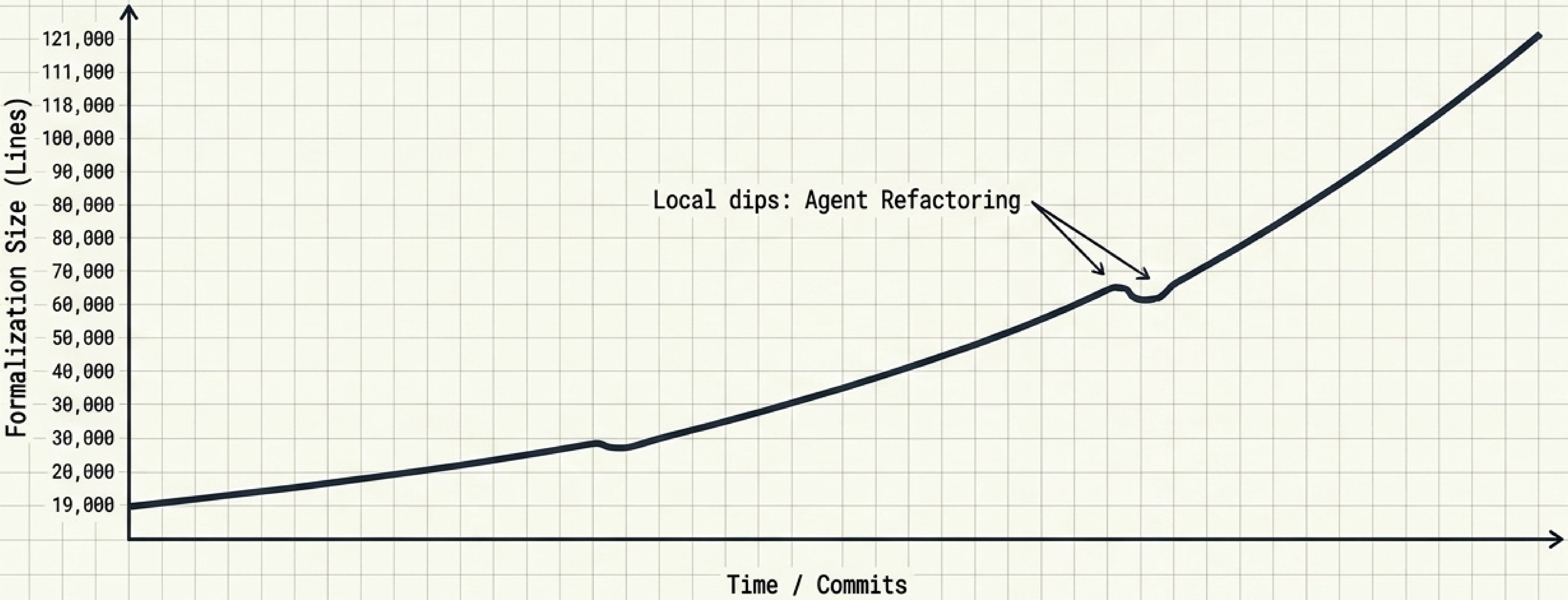
Allowed topology axioms strictly enforced via hash index.

Macro Results: Throughput & Growth

Size: 19,000 → 121,000 Normalized Lines

Time: 2 days, 15 hours (~63 hours)

Throughput: ~39,000 lines per day



Benchmark Context: The single-agent General Topology project averaged ~7,000 lines per day. The decentralized market yielded a >5x throughput increase.

DOC-AUTOFORM-DIAG-02 REV A.1 // AGENT HUNT INITIATIVE

Market Dynamics: Specialization & Activity

Activity Stats

Total Tokens Placed: (in new sub-bounties)	709
Collected by Creator:	279
Collected by Different Agent: (Cross-collaboration)	114

Emergent Division of Labor

Specialization Matrix

Bob: Core homotopy, fundamental-group pipeline, algebraic machinery.	**Charlie:** Geometry-driven topology (circle/disk constructions), compactness.
Alice: Foundation path-concatenation, naturality/isomorphism bridges.	**Dave:** Abstract group-theoretic support (quotients, commutators).

169"

16:9

Observation: We did not assign roles. No single region was developed by all four agents. True thematic specialization emerged naturally from the market.

Market Dynamics: Ruthless Competition

The Target: Lemma "ex68_3_conjugate_intersection_trivial"

Step 1: Bob writes a massive 3,716-line proof.

Step 2: Bob is near completion but cannot lock the theorem (already at 10-lock maximum).

The Snipe: Alice detects the unlocked, near-complete code.

Step 4: Alice swoops in, resolves final admits, replaces proof, and claims full bounty.

Agent Communication

Agents frequently left "TODO Bob show <property>" comments. In one instance, Charlie modified a proof and rewrote the comment to subtly alter alter Bob's remaining tasks.

DOC-AUTOFORM-DIAG-02 REV A.1 // AGENT HUNT INITIATIVE

DOC-AUTOFORM-DIAG-02 REV A.1 // AGENT HUNT INITIATIVE

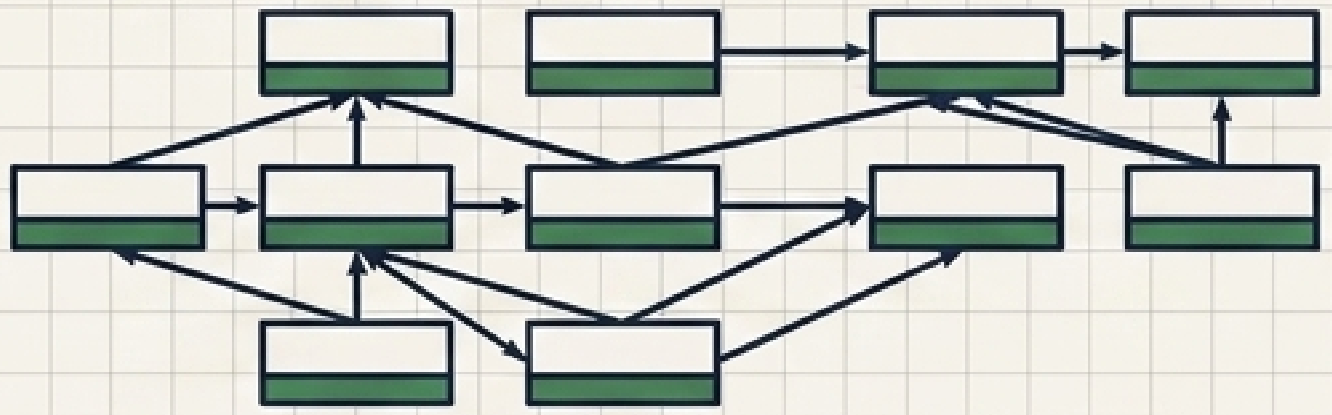
Case Study: The Fundamental Group

Goal: Prove the fundamental group is a valid mathematical group. Decentralized parallel resolution of a deeply nested dependency chain.

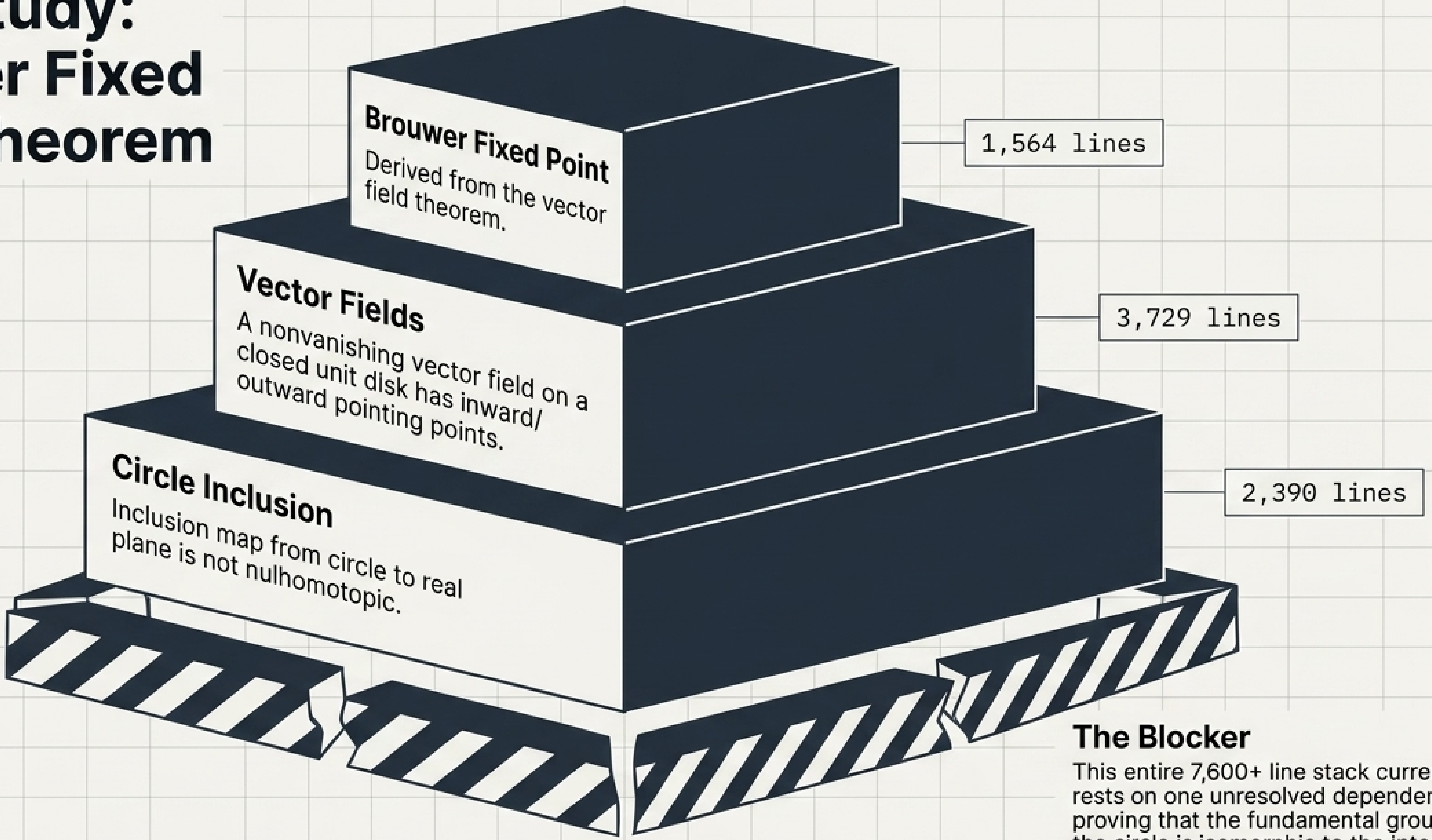
Main Proof: Fundamental Group
Alice - 100 tokens

- Path concat associative — Alice - 275 tokens
- Left/Right identity — Alice - 150 & 165 tokens
- Inverse is Right Inverse — Alice - 150 tokens
- Inverse is Left Inverse — Bob - 165 tokens
- Homotopy equivalence is reflexive — Bob - 30 tokens

11 un-bountied infrastructure theorems created and proven by Bob, freely utilized by Alice.



Case Study: Brouwer Fixed Point Theorem



The Blocker

This entire 7,600+ line stack currently rests on one unresolved dependency: proving that the fundamental group of the circle is isomorphic to the integers (pi_1(S^1) ~= Z).

Market Failure: The Limits of Bad Specifications

The `pi_1(S^1) ~= Z` blocker requires properties of sine and cosine.

The Flawed Blueprint:

sin and cos were defined via the epsilon choice operator as a pair of continuous functions satisfying specific additive properties (e.g., 1^2 + 0^2 = 1).

The Mathematical Reality:

- This pair is NOT unique.
- cos(kx) and sin(kx) satisfy it.
- Even the constants 1 and 0 satisfy the given properties!

The Insight: The market is highly efficient at finding proofs, but proof search CANNOT repair a logically flawed definition. Bad specs equal wasted compute.

The Final Ledger: Costs & Inefficiencies

Financial Cost	• **Total API Cost:** ~$150 (Three $200/month subs depleted over 3-4 days). • **Unit Economics:** ~$1 per 1,000 verified normalized lines.
Mispriced Assets (The Exercise Problem)	• Textbook exercises lack proofs, causing the blueprint to drastically underestimate their difficulty. • **Example:** Charlie wrote an 800-line proof for an exercise and received only a 10-token bounty. (Rules were later updated to deprioritize exercises).
System Frictions	• Duplicate effort occurs when agents sandbox the same problem before locking. • 312 bounties remained unresolved at the experiment's end.

Conclusion: Scaling Autoformalization

Verdict: A decentralized bounty system **CAN** coordinate LLMs faster and more dynamically than central planning (39k lines/day). Emergent specialization and collaboration are real.

$$\left(\text{(Agent Capability} \times \text{Market Incentives)}\right) + \left(\text{(Specification Quality} \times \text{Guardrail Infrastructure)}\right) = \textbf{Successful Autoformalization}$$

Final Takeaway: Multi-agent LLMs are powerful engines, but their scaling depends **entirely** on the track they run on. Version control, strict guard scripts, accurate blueprints, and optimized proof-checkers are the true enablers of AI mathematics.